A third fixed point in a singleton prefix

Human Compression project: L044 continuation

Retain the complete critical singleton hypotheses and the acquired small-support budget. Thus $n = 2r \geq 14$, the actual natural G_0 is $\mathrm{Alt}(\Omega \setminus \{a\})$, full ambient IP holds, and G is a transitive even string action. For $4 \leq g < r$ put

$$P = G_{<g}, \quad Y = \mathrm{supp}(P), \quad C = a^P, \quad H = G_{\{2,\dots,g-1\}}, \quad b = a^{\rho_0}.$$

Assume P is intransitive on Y, $|Y| \leq 2g+1$, and every colour $2, \dots, g-1$ has two ambient-eligible edges.

Theorem 1. *The group H cannot have precisely three fixed points in Y and precisely two nontrivial orbits.*

Proof. Suppose this decomposition holds. The acquired budget gives $|Y| = 2g + 1$, fixed set $\{a, b, z\}$ and two nontrivial orbits U, V. On the fixed set $\rho_0 = (a\ b)$ and fixes z. Since $z \in Y$ and both ρ_0 and H fix it, ρ_1 moves z.

Strict growth gives $|C| \geq g + 1 \geq 5$, so C contains an H-orbit. It cannot contain both: the only possible point left outside would be z, which is moved by P. Thus P has exactly two nontrivial orbits in Y, namely

$$C = \{a, b\} \cup V \cup Z_C, \qquad E = U \cup Z_E, \qquad \{Z_C, Z_E\} = \{\{z\}, \varnothing\}. \tag{1}$$

Both are preserved by ρ_1. The involution ρ_0 centralizes H. It cannot exchange U, V, since then the P-orbit C would contain both. On each preserved orbit it is either free or the identity. It cannot be the identity on both, since it would be the odd permutation $(a\ b)$.

If it is free on both, its support is $Y \setminus \{z\}$. Since ρ_1 moves z, the group $P_1 = \langle \rho_0, \rho_1 \rangle$ has support exactly Y. Every mark of index at least three centralizes P_1 and preserves that support. The mark $\rho_2 \in P$ also preserves Y. Thus Y is a proper nonempty G-invariant set, impossible. Consequently ρ_0 is free on exactly one of U, V. Its evenness makes the size of that orbit twice an odd integer; the acquired exclusion of an internally swapped two-point H-orbit makes its size at least six.

We shall repeatedly use the actual first-prefix stabilizer. Write $C_1 = a^{P_1}$ and $m = |C_1|$. Strict growth gives $m \geq 3$. Actual natural G_0 and full IP imply

$$(P_1)_a = P_1 \cap G_0 = \langle \rho_1 \rangle, \qquad |P_1| = 2m. \tag{2}$$

Hence the product of the two distinct involutions has order m. Every mark of index at least three centralizes P_1, fixes a, and therefore fixes C_1 pointwise.

Put $K = G_{\{3,\dots,g-1\}}$. This is a nonempty interval of nonidentity marks. Its support lies in Y, and its colours have ambient-eligible pairs. If f_K and c_K count its fixed points in Y and its nontrivial orbits, the acquired interval theorem gives

$$2(g-3) \leq |Y| - f_K - c_K, \qquad f_K + c_K \leq 7, \qquad c_K \geq 1. \tag{3}$$

First suppose ρ_0 is free on U and fixes V. The point $x = b^{\rho_1}$ is different from b, lies in C and is fixed by ρ_0. Thus $C_1 = \{a, b, x\}$, with marks $(a\ b)$ and $(b\ x)$. Equation (2) gives actual $P_1 \cong S_3$. On $C \setminus C_1$, the mark ρ_0 is trivial, so the relation $(\rho_0\rho_1)^3 = 1$ also makes ρ_1 trivial there.

If $z \in C$, then $E = U$ is free under ρ_0. Its P_1-orbits have sizes two or six, since their sizes divide six and cannot be odd. They are precisely all the orbits of those sizes in the ambient set. Every mark of index at least three preserves their union E, while ρ_0, ρ_1, ρ_2 do so by (1) and the H-orbit property. This contradicts transitivity.

Suppose instead that $z \in E$. Its P_1-orbit D has size three. Indeed it is nontrivial because ρ_1 moves z; the possible orbit sizes are two, three and six. On a two-point orbit ρ_0 has either zero or two fixed points, and a six-point transitive action of a group of order six is regular. Neither can contain the unique ρ_0-fixed point z in E. Every other orbit in E has size two or six. Thus C_1, D are the only two three-point P_1-orbits globally. Each mark of index at least three fixes C_1 pointwise, hence preserves D. It also fixes D pointwise: it centralizes P_1 and must fix the unique ρ_0-fixed point z in D, after which transitivity propagates fixation.

Consequently $f_K \geq 6$. Equation (3) forces $f_K = 6, c_K = 1$. The sole nontrivial K-orbit is $Y \setminus (C_1 \cup D)$, of odd size $2g - 5$. Since ρ_0 centralizes K, it preserves this unique nontrivial orbit and fixes it pointwise: a centralizing involution on an odd transitive orbit has a fixed point, then fixes the whole orbit. But D contains only two points of U, and $|U| \geq 6$. The remaining points of U lie in that odd orbit and are moved by ρ_0, a contradiction.

Now suppose ρ_0 is free on V and fixes U. If $z \in E$, then $\mathrm{supp}(\rho_0) = C$. The mark ρ_1 preserves C, and every later mark centralizes ρ_0, so C is globally invariant, impossible. We are left with $z \in C$, $E = U$ and

$$\mathrm{supp}(\rho_0) = C \setminus \{z\}, \qquad \rho_0|_E = 1. \tag{4}$$

Both marks preserve C, E, and ρ_1 moves z.

If m is odd, the relation $(\rho_0\rho_1)^m = 1$ and $\rho_0|_E = 1$ imply $\rho_1|_E = 1$. By (4) and the movement of z, the support of P_1 is then exactly C. Every mark of index at least three preserves it by centralization; $\rho_2 \in H$ preserves it by (1). Again G is intransitive.

If m is even, consider $D = z^{P_1} \subseteq C$. The involution ρ_0 has exactly one fixed point on D, so $|D|$ is odd. As ρ_1 moves z, $|D| \geq 3$. Thus $D \neq C_1$. Every other P_1-orbit in C has even size, because ρ_0 is free on it; every orbit in E has size one or two, because ρ_0 is trivial there. Points outside Y are fixed. Therefore D is the unique nontrivial odd-sized P_1-orbit in the whole ambient set. Each mark of index at least three preserves D, fixes its unique ρ_0-fixed point z, and hence fixes D pointwise. It also fixes C_1 pointwise. Since even $m \geq 3$ means $m \geq 4$, we obtain $f_K \geq |C_1| + |D| \geq 7$, contrary to (3). All cases are excluded. □

Attribution and remaining scope. The critical singleton setting, strict prefix growth, ambient interval bound and internal two-point exclusion are acquired. The complete support split, the actual first-prefix orbit argument, and the use of six or seven fixed points to settle the third fixed point are this continuation. The source context is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, Propositions 5.28–5.32. Together with the separately accepted $f = 2, c = 2$ exclusion, this leaves only $f = 2, c = 3$ in the stated small-support budget. No projected IP, general faithful orbit action, or finite census is assumed. Complete singleton routing and the other principal L044 obligations remain open within this recovery; no whole-original or formal credit is added.